



Lenz's Law

Lenz's Law: To find the direction of induced emf or current in the circuit Lenz's law is used.

Statement: "The direction of the induced current (or the emf) in the circuit will be such so as to oppose the very cause due to which it is produced"

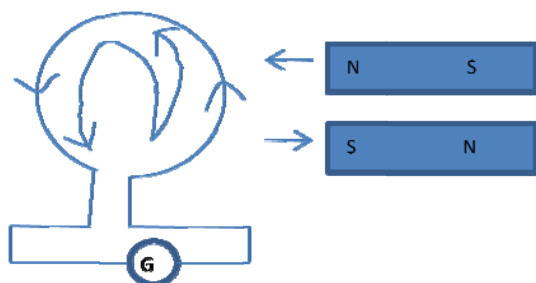


Fig-1

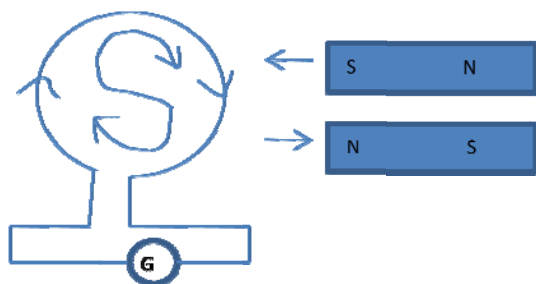


Fig-2

In Fig(1) the cause of getting induced current is north pole of the bar magnet is approaching towards the coil (or south pole of the bar magnet is receding away from the coil). The induced current flowing through the coil will produce a magnetic field which will oppose the cause i.e. Oppose the approaching N-pole (or receding South Pole).

This will be possible only if magnetic field produced by the induced current is along opposite to the direction of the approaching North Pole.

If we place the thumb on the right hand in the direction of magnetic field the curl of the finger tips gives the direction of the induced current flowing through the circuit.